

MACHINE REPRESENTATION OF PROGRAMS

Computers execute machine code

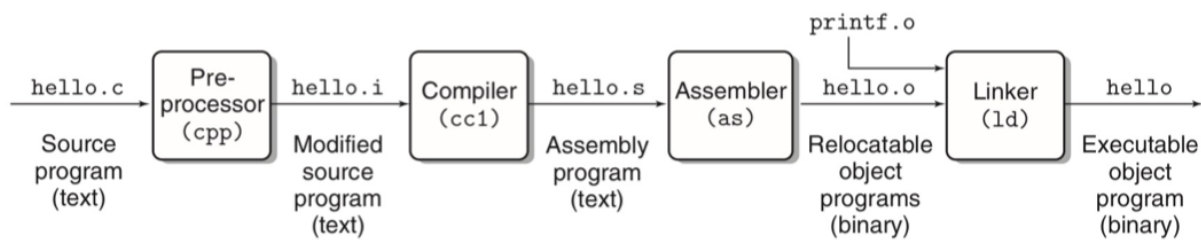
↳ sequence of bytes encoding low level operations



A compiler generates machine code by-

- Rules of the programming language
- Instruction set of the target machine
- Conventions of the Operating System

→ The Compilation Process



Compiler Optimizations can rearrange execution order, eliminate unwanted computations, replace slow operations and change recursive computation to iterative ones.

Moore's Law - The number of transistors per chip will double every year.

↳ Moore's Law has sadly ended. I believe Physics caught up.

The format and behavior of machine level programs is defined by the instruction set architecture. Most ISAs describe the behavior of the program as if each instruction is executed in sequence, with one instruction completing before the next one begins.